ARYAN PARAB

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EDUCATION

University of Southern California Los Angeles, CA
M.S. in Computer Science | GPA: 3.85/4.0 January 2025 - December 2026
Coursework: Coursework: Algorithms, Operating Systems, Database Architecture, Web Technologies, Information Retrieval, Robotics
Thadomal Shahani Engineering College Mumbai, India
Bachelor of Engineering, Computer Engineering | GPA: 9.1/10 August 2019 - May 2023

TECHNICAL SKILLS

Languages: Python, C++, C, Java, TypeScript/JavaScript, SQL
Robotics & Learning: PyTorch, Imitation Learning, Behavioral Cloning, DAgger, Reinforcement Learning, OpenAI Gym, MuJoCo
Systems: Linux/Unix, Processes, Threads, Scheduling, Synchronization, File Systems, System Calls, Concurrency
Backend & Tools: FastAPI, Flask, PostgreSQL, MongoDB, Docker, AWS EC2/S3, Git, Unity
ML: TensorFlow/Keras, scikit-learn, Deep Learning, Computer Vision fundamentals, NumPy

EXPERIENCE

Cadence Design Systems San Jose, CA
Generative AI Software Engineering Intern May 2026 – Present
-Architected an autonomous execution system that converts structured QA specifications into multi-step workflows involving planning, simulation, execution, debugging, and recovery.
-Designed dependency-aware planning across 1,000+ test cases, decomposing complex tasks into reusable execution components and shared state dependencies.
-Integrated Playwright-based simulation with external automation tools, enabling the system to observe runtime state, execute actions, identify failures, and retry failed steps.
-Built a daily traffic-analysis pipeline that preprocesses and clusters security events using JA4 fingerprints, IP behavior, and request characteristics before downstream automated analysis.
-Developed automated decision workflows that compare observed system behavior against existing rules, detect failure conditions, and recommend corrective actions.
-Added scheduled execution and recovery mechanisms to support reliable recurring operation with minimal manual intervention.

USC Marshall School of Business — Los Angeles, CA
Research Assistant **February 2026 – Present**
-Developed Python infrastructure for running and evaluating repeated experiments across multiple model configurations.
-Built automated benchmarking and validation pipelines for comparing system behavior and identifying failure modes.
-Improved experimental reproducibility through reusable evaluation, aggregation, and data-validation workflows.

PROJECTS

Imitation Learning for Robotic Control
Python, PyTorch, MuJoCo, OpenAI Gym, NumPy November 2025
-Implemented Behavioral Cloning and DAgger for continuous-control robotic environments using expert demonstrations and iterative on-policy data collection.
-Trained and evaluated control policies on HalfCheetah-v4 and Hopper-v4, comparing imitation against iterative dataset aggregation.
-Improved HalfCheetah performance from 3825.7 to 4025.8 average return, reaching 99.8% of expert performance after DAgger.
-Improved Hopper performance from 891.4 to 3714.5 average return, reaching 99.9% of expert performance and a 316.7% improvement over Behavioral Cloning.
-Analyzed distribution shift between expert demonstrations and learned-policy trajectories, demonstrating how DAgger mitigates compounding errors in sequential control.

Weenix Operating System Kernel — Systems Programming
C, Linux, GDB, Concurrency December 2025
-Implemented Unix-like kernel functionality across process/thread management, scheduling, synchronization, VFS operations, and system calls within a 10,000+ line C codebase.
-Implemented process lifecycle functionality including fork(), exec(), waitpid(), scheduling, and thread synchronization.
-Built and debugged filesystem functionality including file operations, directory management, path resolution, and VFS abstractions.
-Diagnosed race conditions, process lifecycle bugs, deadlocks, and filesystem failures using GDB, kernel logs
-Validated concurrent execution across workloads containing 100+ processes.

LevelDB BitWeaving & ZoneMap Optimization
C++, LevelDB, LSM Trees, SIMD, Storage Systems April 2026
Modified LevelDB's storage engine to add per-block metadata and selective filtering before disk I/O, reducing unnecessary block reads during range queries.Benchmarked the implementation across datasets up to hundreds of millions of records and achieved up to 31.2× query speedup with 92.3% of blocks skipped on selective workloads.